Shenghan **Gao**

TELECOMMUNICATIONS · INTEGRATED PHOTONICS · COHERENT OPTICAL SYSTEMS

(+1) 585-489-1187 | shg108@pitt.edu | gsh1994 | sheng-han-gao

Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in coherent optical links, semiconductor-laser injection locking, and integrated photonic chip design. Dissertation work connects telecom-relevant carrier recovery, modulation-transfer analysis, fiber coherent-interference experiments, and chip-scale coherent processing nodes. Experienced with optical testbeds, lasers, EOMs, OSAs, SVAs, photodetectors, balanced detection, PID phase stabilization, KLayout photonic layouts, Lumerical/COMSOL/Zemax simulation, and Python/MATLAB data analysis. Targeting telecommunications, silicon photonics, coherent optics, and photonic hardware R&D roles.

Core Skills

Telecom optics	Coherent optical links, carrier recovery, optical injection locking, EOM modulation, sideband/RF analysis, balanced detection
Photonics chips	KLayout, waveguides, grating couplers, splitters/combiners, interferometric paths, e-beam lithography, SEM, optical characterization
Lab systems	Tunable lasers, slave/master laser setups, OSAs, SVAs, photodetectors, lock-in amplifiers, PID control, fiber stretchers, VOAs, PCs
Simulation	Semiconductor-laser rate equations, injection-ratio/detuning sweeps, modulation transfer, Lumerical, COMSOL, Zemax, CodeV
Software	Python, NumPy, SciPy, Pandas, Matplotlib, PyVISA/SCPI, NI MAX, MATLAB, C++, Git/GitHub, pytest
ML / CV	Machine learning coursework, PyTorch, TensorFlow, scikit-learn, OpenCV, CNN image recognition

Dissertation and Research Experience

University of Pittsburgh
Pittsburgh, PA

PH.D. RESEARCH ASSISTANT | ADVISOR: NATHAN YOUNGBLOOD — *Sep 2019 - present*

- Developed a dissertation framework for remote coherent optical processing using optical injection locking, targeting systems where remotely generated optical data must be processed without repeated O/E conversion, digitization, buffering, and re-transmission.
- Modeled semiconductor-laser injection locking with rate equations, sweeping injection ratio, detuning, linewidth-enhancement factor, and operating threshold to map stable carrier-locking regions using phase- and intensity-stability criteria.
- Analyzed AM/PM modulation transfer through injection-locked lasers, extracting AM-to-AM, AM-to-PM, PM-to-AM, and PM-to-PM response channels to identify operating regimes that recover a clean coherent carrier while suppressing unwanted remote sideband transfer.
- Built end-to-end coherent multiply-and-accumulate simulations using remote/local symbol sequences, interference, balanced photodetection, and temporal integration; showed that lower injection ratios can improve computational fidelity when locking margin remains sufficient.
- Designed and executed fiber injection-locking experiments with tunable master/slave lasers, VOAs, polarization control, OSAs, photodetectors, and SVA measurements; matched -15 dB experimental injection behavior to calibrated simulations using $\alpha = 2$ and threshold 1.944.
- Implemented coherent fiber interference validation using an injection-locked local carrier, retained remote signal branch, lock-in detection, PID phase feedback, and fiber-stretcher control to stabilize phase-sensitive measurements.
- Designed chip-scale coherent-processing layouts with grating-coupler access, waveguide routing, splitting/combining structures, and interferometric paths to translate fiber-validated injection-locking workflows onto integrated photonic platforms.

University of Rochester
Rochester, NY

RESEARCH ASSISTANT | ADVISOR: XI-CHENG ZHANG — *Sep 2017 - Aug 2019*

- Generated THz signals from liquid plasmas and analyzed how temperature, polarity, and viscosity affected emission strength and spectral content.
- Processed time- and frequency-domain experimental data with SNR, repeatability, sensitivity, and regression analysis; prototyped digital cloaking using lenslet arrays and MATLAB pixel remapping.

Industry Experience

Lumentum Operations LLC
San Jose, CA

RESEARCH INTERN | MENTOR: JOE WEN — *May 2023 - Aug 2023*

- Built MATLAB/Python workflows for FIR-filter equalization of communication signals distorted by intersymbol interference, connecting signal-processing design to recovered waveform quality.
- Collected measured signal data and applied FFT-based frequency analysis to compare filter responses, tune coefficients, and improve demodulated-signal recovery.

Sunny Optics
Ningbo, China

RESEARCH INTERN | MENTOR: CHAO XU — *May 2018 - Aug 2018*

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance and packaging requirements.
- Worked with opto-mechanical engineers through lab integration, using test feedback to refine alignment and system functionality.

Education

University of Pittsburgh
Pittsburgh, PA

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING — *Dec. 2026 expected*

Georgia Institute of Technology — *Atlanta, GA*

M.S. in Computer Science, Machine Learning track — *May 2026 expected*

University of Rochester — *Rochester, NY*

M.S. in Optics (thesis track) — *Aug. 2019*

B.S. in Optical Engineering — *May 2017*

B.S. in Applied Mathematics — *May 2017*

- Relevant training: optical communications, silicon photonics simulation, signal processing, software engineering, deep learning, circuit design, lens design, coating design, photonic device design, and computational imaging.

Selected Publications and Talks

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., K. Lüdge, F. Da Ros, N. Youngblood, submitted to APL, under review, 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., Invited PhD Candidate Talk, Pittsburgh Quantum Institute, Pittsburgh, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, SPIE, 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. Jin, Y. E., **S. Gao**, X.-C. Zhang, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. Gao, Q. Jin, Y. E., X.-C. Zhang, SPIE, 2019